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Agent-Based Micro-Modeling of Epidemic Dynamics

Integration with Macro-Level Model

Predictive Control

for Multi-City Epidemic Management

Bachelor's Thesis - Electrical Engineering - April 2026

```
func (m *Manager) NewConfiguration(opts ...gorums.ConfigOption) (c *Configuration, err error) {
    if len(opts) < 1 || len(opts) > 2 {
        return nil, fmt.Errorf("wrong number of options: %d", len(opts))
    }
    c = &Configuration{}
    for _, opt := range opts {
        switch v := opt.(type) {
        case gorums.NodeListOption:
            c.Configuration, err = gorums.NewConfiguration(m.Manager, v)
            if err != nil {
                return nil, err
            }
        case QuorumSpec:
            // Must be last since v may match QuorumSpec if it is interface{}
            c.qspec = v
        default:
            return nil, fmt.Errorf("unknown option type: %v", v)
        }
    }
    // return an error if the QuorumSpec interface is not empty and no impl
    var test interface{} = struct{}{}
    if _, empty := test.(QuorumSpec); !empty && c.qspec == nil {
        return nil, fmt.Errorf("missing required QuorumSpec")
    }
    return c, nil
}
```

I, **Mohamed Ahmed Jama & Håkon rullestad håvarstein** , declare that this thesis titled, “Agent-Based Micro-Modeling of Epidemic Dynamics Integration with Macro-Level Model Predictive Control for Multi-City Epidemic Management” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a bachelor's degree at the University of Stavanger.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.

“Programming is a nice break from thinking.”

– Leslie Lamport

Abstract

This thesis presents the design, implementation, and simulation-based evaluation of an agent-based (micro-level) epidemic model integrated with a macro-level Model Predictive Control (MPC) framework for multi-city epidemic management. The macro model, developed by a collaborating student, captures population-level compartmental dynamics using an extended SIRVHDB (Susceptible-Infected-Recovered-Vaccinated-Hospitalized-Dead- Breakthrough) formulation and drives epidemic intervention through MPC. The present work focuses on the micro-level component: a stochastic, agent-based simulation environment implemented in MATLAB that models the explicit movement, contact, infection, and disease progression of individual agents across two interconnected cities.

The micro-model receives time-varying transmission rate signals $\beta_1(t)$ and $\beta_2(t)$ generated by the macro-level MPC controller and reproduces realistic epidemic trajectories at the individual level. Key design features include proximity-based infection with spatial distance weighting, inter-city commuting dynamics, vaccination with breakthrough immunity, hospitalization logic with capacity tracking, and a comprehensive real-time visualization dashboard. Simulation results demonstrate that MPC-generated transmission reductions successfully suppress infection peaks in both cities, that hospital load remains within capacity thresholds under MPC feedback, and that inter-city mobility creates cross-city epidemic coupling that neither a purely macro nor a purely micro model alone can adequately capture.

The findings support the value of multi-scale modeling in epidemic control: the macro model provides computationally efficient, provably stabilizing control signals, while the micro model reveals heterogeneous, stochastic dynamics that aggregate models cannot represent. Together, they constitute a coupled simulation framework with direct relevance to data-driven public health decision-making.

Acknowledgements

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I also thank my Håkon, with whom I jointly developed the macro-level SIRVHDB model that forms the foundation of this thesis. The collaborative effort made it possible to build a coherent two-part study. Any errors that remain are my own.

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Symbols and Abbreviations

Roman Symbols

Symbol	Meaning	Units	First section
b	Breakthrough reinfection probability (SIRVB model)	dimensionless	Sec. 3.1
d_{ij}	Euclidean distance between agents i and j	spatial units	Sec. 5.4.2
D_c	Dead compartment, city c	agents	Sec. 1.4
E	Exposed compartment (SEIR model)	agents	Sec. 3.1
h	Daily hospitalisation probability (unvaccinated infected), $h = 0.05$	dimensionless	Sec. 3.6
h_v	Daily hospitalisation probability (vaccinated breakthrough infected), $h_v = 0.02$	dimensionless	Sec. 5.5
H_c	Hospitalised compartment, city c	agents	Sec. 1.4
$H_{\text{cap},c}$	Hospital bed capacity, city c	agents	Sec. 4.2
$H_{\text{load}}(t)$	Combined hospital load ratio $= (H_1 + H_2) / (H_{\text{cap},1} + H_{\text{cap},2})$	dimensionless	Sec. 4.2
H_{ref}	MPC hospital load reference threshold (= 0.8)	dimensionless	Sec. 4.2
I_c	Infected (symptomatic, unvaccinated) compartment, city c	agents	Sec. 1.4
$I_{v,c}$	Vaccinated-infected (breakthrough) compartment, city c	agents	Sec. 1.4
$I_{0,c}$	Initial infected seed count, city c	agents	Sec. 5.3

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Symbol	Meaning	Units	First section
$\mathcal{I}_c(t)$	Total infectious agents in city c : $I_c(t) + I_{v,c}(t)$	agents	Sec. 5.4
k	Maximum MPC intervention effectiveness ($k = 0.70$)	dimensionless	Sec. 4.1
M	Inter-city travel/contact matrix (2×2)	dimensionless	Sec. 4.1
$M_{cc'}$	Element of M : fraction of contacts in city c with city c'	dimensionless	Sec. 4.1
N	Total agent population, $N = N_1 + N_2$	agents	Sec. 5.3
N_c	Population of city c ; baseline $N_1 = N_2 = 500$	agents	Sec. 4.1
$N_c^{\text{alive}}(t)$	Non-dead agents in city c at time t	agents	Sec. 5.4
p_{commute}	Daily inter-city commute probability (= 0.015)	dimensionless	Sec. 5.6
$p_{\text{inf}}^{(S)}(t)$	Daily infection probability for a susceptible agent (mean-field)	dimensionless	Sec. 5.4
$p_{\text{inf}}^{(V)}(t)$	Daily infection probability for a vaccinated agent (mean-field)	dimensionless	Sec. 5.4
$p_{\text{inf}}(i \rightarrow j, t)$	Proximity-based infection probability from agent i to agent j	dimensionless	Sec. 5.4.2
R_0	Basic reproduction number, $R_0 = \beta/\gamma$	dimensionless	Sec. 3.1
R_c	Recovered compartment, city c	agents	Sec. 1.4
R_t	Time-varying effective reproduction number	dimensionless	Sec. 3.1
r_{inf}	Interaction radius for proximity-based infection (= 2.8 spatial units)	spatial units	Sec. 5.4.2
S_c	Susceptible compartment, city c	agents	Sec. 1.4
T	Simulation horizon ($T = 150$ days)	days	Sec. 1.3
$u_c(t)$	MPC intervention intensity for city c ; $u_c \in [0, 1]$, where 1 corresponds to maximum social-distancing enforcement	dimensionless	Sec. 4.1
V_c	Vaccinated-susceptible compartment, city c	agents	Sec. 1.4

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Symbol	Meaning	Units	First section
$V_{0,c}$	Initial vaccinated count, city c	agents	Sec. 5.3

Greek Symbols

Symbol	Meaning	Units	First section
α	Contact-intensity scaling factor for proximity infection ($\alpha = 0.35$)	dimensionless	Sec. 4.5
β	Transmission rate (per contact, per day)	day ⁻¹	Sec. 3.1
$\beta_c(t)$	MPC-modulated transmission rate in city c at time t	day ⁻¹	Sec. 4.1
$\beta_{c,\text{nom}}$	Nominal (uncontrolled) transmission rate in city c ($= 0.45$)	day ⁻¹	Sec. 4.1
γ	Recovery rate, unvaccinated infected ($= 1/12$)	day ⁻¹	Sec. 3.1
γ_h	Hospital discharge (recovery) rate ($= 1/25$)	day ⁻¹	Sec. 5.5
γ_v	Recovery rate, vaccinated breakthrough infected ($= 1/9$)	day ⁻¹	Sec. 5.5
η	Vaccine breakthrough factor: relative infection risk for vaccinated agents ($\eta = 0.60$); <i>Note: in Sec. 3.1, η^{-1} denotes the SEIR incubation period following standard notation. The present model uses η exclusively for vaccination.</i>	dimensionless	Sec. 3.6
κ_j	Vulnerability modifier for agent j : $\kappa_j = 0.35$ if vaccinated, 1 otherwise	dimensionless	Sec. 5.4.2
μ	Daily fatality rate, unvaccinated infected ($= 0.001$)	day ⁻¹	Sec. 3.6
μ_h	Daily fatality rate, hospitalised ($= 0.02$)	day ⁻¹	Sec. 5.5
μ_v	Daily fatality rate, vaccinated breakthrough infected ($= 0.0005$)	day ⁻¹	Sec. 5.5

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(Greek symbols, continued)

Symbol	Meaning	Units	First section
ν_c	Daily vaccination rate, city c ($= 0$ in baseline)	day^{-1}	Sec. 5.5
ω_R	Waning immunity rate after recovery ($= 1/110$)	day^{-1}	Sec. 4.1
ω_V	Waning immunity rate, vaccine-induced ($= 1/140$)	day^{-1}	Sec. 4.1

Abbreviations

Abbreviation	Meaning	First section
ABM	Agent-Based Model	Sec. 2.3
COVID-19	Coronavirus Disease 2019	Sec. 2.3
GA	Genetic Algorithm	Sec. 7.3 [†]
MATLAB	Matrix Laboratory (The MathWorks, Inc.)	Sec. 1.3
MPC	Model Predictive Control	Sec. 2.3
ODE	Ordinary Differential Equation	Sec. 2.3
PINN	Physics-Informed Neural Network	Sec. 3.4
QP	Quadratic Program (the convex optimisation problem solved by the MPC at each step)	Pending [†]
SEIR	Susceptible–Exposed–Infected–Recovered (compartmental model)	Sec. 3.1
SIR	Susceptible–Infected–Removed (compartmental model)	Sec. 3.1
SIRV	Susceptible–Infected–Recovered–Vaccinated (compartmental model)	Sec. 3.1
SIRVB	Susceptible–Infected–Recovered–Vaccinated–Breakthrough	Sec. 3.1
SIRVHDB	Susceptible–Infected–Recovered–Vaccinated–Hospitalised–Dead–Breakthrough (macro model used in this thesis)	Sec. 2.3
sMAPE	Symmetric Mean Absolute Percentage Error (GA fitness metric)	Pending [†]
WHO	World Health Organization	Sec. 3.1

[†] Symbol or abbreviation first used in the GA/results chapter (`chapter_micro_abm_and_results.tex`) which is pending final integration (see Prompt 2). The entry is included here for completeness; the section reference will be updated once that chapter is compiled.

Chapter 1

Introduction

1.1 Motivation and Context

The COVID-19 pandemic demonstrated with stark clarity that infectious disease dynamics operate across multiple scales simultaneously [1, 2]. At the population level, aggregate transmission rates determine whether an epidemic grows or recedes. At the individual level, the specific contact patterns, mobility behavior, vaccination status, and comorbidities of each person determine who becomes infected, who is hospitalized, and who dies [3]. Control strategies designed using only population-level models may be adequate on average yet blind to the heterogeneity that drives localized outbreaks, hospital overcrowding in specific regions, or the amplifying effect of high-mobility individuals on cross-community spread.

Control theory has emerged as a natural language for epidemic management. Feedback-based strategies, and Model Predictive Control (MPC) in particular, have been applied to epidemic models to design intervention policies that respect constraints such as hospital capacity limits while minimizing societal disruption. Research has shown that properly designed MPC schemes can drive epidemic compartmental models to disease-free equilibria with formal guarantees of recursive feasibility and stability [4]. However, these guarantees are derived for population-level ordinary differential equation (ODE) models, and it remains important to validate that MPC-generated control signals also produce meaningful epidemic suppression when applied to stochastic, agent-based representations of the same population.

This thesis is the micro-level component of a two-part bachelor thesis project. The macro-level component, developed by a collaborating student, implements a multi-city SIRVHDB compartmental model controlled by MPC. The present

work implements the micro-level counterpart: a spatially explicit, stochastic agent-based model (ABM) that receives the time-varying transmission rate signals produced by the MPC controller and simulates individual-level epidemic dynamics across two interconnected cities. The ABM serves as both a validation tool for the macro model and a richer representation of epidemic heterogeneity [5, 6].

1.2 Problem Statement

The central problem addressed in this thesis is: given that an MPC controller has been designed at the macro (population) level to regulate epidemic transmission rates, does the controlled transmission signal also effectively suppress epidemic dynamics when simulated at the individual (agent) level, accounting for stochastic contact processes, spatial heterogeneity, inter-city mobility, and individual variation in disease progression?

A secondary problem is the design of the agent-based model itself: what level of complexity is necessary and sufficient to capture the epidemic behaviors that matter for evaluating MPC performance, and how should micro-model outputs be aggregated into the feedback signals that the macro model requires?

1.3 Scope and Delimitations

This thesis focuses exclusively on the micro-level simulation component. The macro ODE model, MPC formulation, and Simulink implementation are the responsibility of the collaborating student and are described here only to the extent necessary to understand the interface between the two model levels. The agent-based model is implemented in MATLAB and receives $\beta(t)$ signals from the macro Simulink model through a defined integration interface. No closed-loop feedback from the micro model to the MPC controller is implemented; the coupling is one-directional in this project.

The simulation population consists of 1 000 agents (500 per city) and is run over 150 simulated days. Hospital capacities, vaccination rates, and disease parameters are calibrated to be epidemiologically plausible but are not fitted to any specific real-world outbreak. The visual simulation component (`run_micro_sim_visual.m`) is a separate, display-oriented implementation used to illustrate the dynamics; quantitative results are drawn from the non-

`visual run_micro_sim.m`.

1.4 Contributions

The specific contributions of this thesis are:

- Design and MATLAB implementation of a two-city, spatially explicit stochastic agent-based epidemic model with seven disease states: Susceptible (S), Infected (I), Vaccinated-Infected (I_v), Recovered (R), Vaccinated (V), Hospitalized (H), and Dead (D).
- Implementation of inter-city commuting dynamics that generate realistic cross-city epidemic coupling without requiring full spatially explicit geography.
- Design of an integration interface that accepts time-series $\beta(t)$ signals from the macro Simulink/MPC model and translates them into per-step infection probabilities for each agent.
- A simulation-based evaluation comparing epidemic trajectories under MPC-controlled and uncontrolled transmission, demonstrating the effectiveness of the MPC signal at the micro level.
- A real-time animated visualization dashboard (`run_micro_sim_visual.m`) illustrating agent-level dynamics across both cities simultaneously with a live statistics panel.

1.5 Thesis Outline

Chapter 3 reviews the relevant literature on compartmental epidemic models, agent-based epidemic models, and MPC applied to epidemic control. Chapter 4 describes the macro-level model (developed by the collaborating student) and the macro–micro integration architecture. Chapter 5 presents the design and implementation of the micro-level agent-based model in detail. Chapter 6 presents the simulation results. Chapter 7 discusses the findings, limitations, and directions for future work. Chapter 8 concludes the thesis.

Chapter 2

Introduction

2.1 Questions

- Should the theory hold a different form. More history-like as several of the articles we have read?
- Should the theory be more in the shortened style as in the theory section, or more in the style of the earlier version i have added in appendix?
- Should the tables of variables also hold the values?
- Where to place the model equation?
- Are the figures ok?

2.2 Known issues/errors

- Must place citations
- Must find out how to add the sources to the document
- Full model equation is missing transport and commute
- Values for the MPC are not included, should they?
- Side-by-side figures looks small, the colors on them are also the hardest to see on a black background
- Both our documents must be combined to one, overlapping content must be removed

Page budget for Introduction: 3-5 pages.

2.3 Background and Motivation

- Awaken the reader's interest and convince her why the theme is important.
- Background information might be historical in nature, or it might refer to previous research or practical considerations.
- It should be written on a level that it's understandable by anyone with a computer science bachelor's degree.

2.4 Objectives

- Define the goals of your study.
- It might be presented as a bullet list.

2.5 Approach and Contributions

- Give a brief summary of your overall approach.
- Summarize the specific contributions that you made in this thesis (implementation, empirical results, analysis, etc.).
- It might be presented as a bullet list.

2.6 Outline

- Give an overview of the main points and the structure of your thesis.
- Examples: "Chapter 2 covers ... Chapter 3 describes ..."
- Show how the different parts (chapters) relate to each other.

Chapter 3

Background and Related Work

3.1 Compartmental Epidemic Models

The mathematical modeling of infectious diseases traces back to Kermack and McKendrick’s foundational SIR (Susceptible–Infected–Removed) model introduced in 1927 [7]. In the SIR framework, the population is partitioned into three compartments, and the dynamics are governed by a system of ordinary differential equations parameterized by a transmission rate β and a recovery rate γ . The ratio $R_0 = \beta/\gamma$, known as the basic reproduction number, determines whether an epidemic grows ($R_0 > 1$) or decays ($R_0 < 1$). The mathematical properties of this threshold condition are thoroughly reviewed in Hethcote [8], and early epidemic growth phases are characterized in detail in Chowell et al. [9].

Numerous extensions of the SIR model have been developed to capture additional epidemiological phenomena [8]. The SEIR model adds an Exposed (E) compartment representing individuals who have been infected but are not yet infectious, with incubation period η^{-1} . The dynamics of the SEIR model during the COVID-19 pandemic are studied in He et al. [10], who characterize the sensitivity of epidemic trajectories to β and demonstrate the importance of accurate parameter estimation for reliable forecasting. The SIRD model incorporates vaccination, and the SIRDV model, studied in the context of COVID-19, adds a breakthrough infection pathway allowing recovered or vaccinated individuals to be reinfected with probability b . The macro model in the present project extends this to SIRDVHDB, adding Hospitalized (H) and Dead (D) compartments with city-specific dynamics.

Ariyaratne et al. [11] applied the SIRVB model to COVID-19 data from the World Health Organization across 242 countries, demonstrating that the breakthrough parameter b crucially determines the equilibrium reproduction number R_t in the post-pandemic phase and that SIRVB outperforms both SIR and SIRV in explaining late-pandemic dynamics. Their work confirms that the simplest model including both vaccination and breakthrough effects is sufficient to capture the main kinetic features of a multi-year pandemic. Epidemic parameter estimation across countries was further enabled by the Johns Hopkins COVID-19 dashboard, which aggregated and curated global case data throughout the pandemic period [1].

3.2 Agent-Based Models in Epidemiology

Agent-based models (ABMs) represent an alternative and complementary approach to epidemic modeling. Rather than tracking aggregate compartment populations, ABMs explicitly simulate the state, behavior, and interactions of each individual (agent) in the population. Agents are characterized by attributes such as disease state, location, vaccination status, and behavioral rules such as movement patterns and contact rates [5]. A comprehensive recent review of ABM approaches, applications, and future directions is provided by Zhang et al. [5], who survey models ranging from simple lattice-based contact simulators to large-scale digital-twin city representations.

ABMs offer several advantages over compartmental models. They naturally accommodate population heterogeneity, including age structure, comorbidities, and behavioral variation. They can incorporate spatial structure and explicit contact networks, capturing the role of superspreaders, geographic clustering, and mobility corridors. They produce stochastic outputs that reflect the inherent randomness of disease transmission at the individual level, which is especially important for small populations or in the early stages of an outbreak when the law of large numbers does not apply. The importance of spatial spread in epidemic ABMs was demonstrated by Merler et al. [12] in the context of the 2014 Ebola outbreak in Liberia, where an agent-based model capturing spatiotemporal heterogeneity explained the effectiveness of ring vaccination and isolation policies that purely aggregate models failed to predict.

A landmark open-source ABM implementation for COVID-19 is Covasim [13], which represents contacts through layered networks (household, school, workplace, community) and has been calibrated to real outbreak data in multiple

countries. The present project uses a simpler two-city spatial model, but the Covasim experience motivates the use of agent-level randomness and structured mixing as key features. Contact matrices inferred from survey data [3] provide empirical support for the mean-field infection probability model used in the non-visual simulation.

The tradeoff is computational cost. While a compartmental ODE model can be integrated in milliseconds, an ABM with millions of agents may require hours of computation. For this reason, ABMs are typically used for detailed regional analysis or model validation rather than for rapid optimization or control design [6]. The approach taken in the present project reflects this tradeoff: the MPC controller operates on the computationally efficient macro model, while the ABM serves as a higher-fidelity simulation environment for evaluation. Comparing non-pharmaceutical interventions across different epidemic scenarios illustrates that the choice of model resolution significantly affects policy conclusions [14].

3.3 Multi-Scale Epidemic Modeling

Multi-scale models couple epidemic dynamics at different levels of resolution. In the context of this project, the macro level represents population dynamics across two cities using compartmental ODEs, while the micro level represents individual agents within each city. The key interface is the transmission rate $\beta(t)$: the macro model computes a population-level effective transmission rate that is then used to parameterize individual infection probabilities in the ABM.

This coupling is consistent with the mean-field approximation underlying compartmental models: the per-contact infection probability in the ABM, when averaged over a well-mixed population, should recover the macro-level transmission dynamics. Deviations from the mean field - due to spatial clustering, behavioral heterogeneity, or stochasticity - are precisely what the micro model is designed to reveal [15].

3.4 Model Predictive Control for Epidemic Management

Model Predictive Control (MPC) is an optimization-based control strategy in which a finite-horizon optimal control problem is solved repeatedly online. At

each time step, the controller receives the current state, solves an optimization problem over a prediction horizon, implements the first portion of the resulting control trajectory, and then repeats. MPC naturally handles state and input constraints, making it well suited to epidemic management, where constraints such as maximum hospital occupancy and minimum intervention levels are directly relevant [16, 17].

Esterhuizen et al. [4] present a rigorous MPC formulation for the SEIR model without terminal ingredients. They prove recursive feasibility and asymptotic convergence to disease-free equilibria from any initial state in the admissible set, provided the prediction horizon is sufficiently long. Köhler et al. [18] develop a robust and optimal predictive control framework for COVID-19, demonstrating that constraint-aware MPC can effectively bound infection and hospitalization curves even under model uncertainty. Sauerteig et al. [19] further tailor MPC design to compartmental epidemic models and validate it on realistic parameter sets.

Multi-region MPC formulations are particularly relevant to the present two-city framework. Carli et al. [20] design MPC policies for a multi-region epidemic scenario and study how intervention coordination across regions reduces total burden compared to independent local policies. Similarly, Tsay et al. [21] formulate a state-estimation and optimal control problem for COVID-19 across US regions, coupling data-driven β estimation with MPC feedback. Nonlinear MPC formulations with logic constraints for epidemic management are studied in [22], while stability and optimal control strategies for COVID-19 epidemic models are analysed in [23]. Nonlinear control of infection spread using a deterministic SEIR model is addressed in [24].

Recent work has extended MPC to more exotic plant models. Zhong et al. [25] combine physics-informed neural networks (PINNs) with MPC to handle partially unknown SIR dynamics, achieving data-efficient control without full model knowledge. Kosyanov et al. [26] apply MPC to the SIIR model (with two infected classes distinguishing symptomatic and asymptomatic) and compare performance across different prediction horizons. Sayed Hassen et al. [27] design a data-driven MPC for SARS-CoV-2 in Mauritius, demonstrating the applicability of the framework to island epidemic scenarios with constrained healthcare capacity.

The macro model in the present project uses a similar MPC formulation applied to the extended SIRVHDB model with two cities and a travel matrix, with hospital load H/H_{cap} as the controlled output. The MPC generates time series $\beta_1(t)$ and $\beta_2(t)$ representing the effective transmission rates in each city

under the prescribed intervention policy [28]. A mean field perspective on epidemic modeling and control that connects the population-level ODE formulation to the individual-level ABM is provided by Roy [29].

3.5 Hybrid and Data-Driven Approaches

A growing strand of the literature combines the computational efficiency of compartmental models with the behavioral richness of agent-based simulation. Polcz et al. [6] develop a hybrid framework that runs MPC on an ODE model for fast online optimization, then evaluates candidate policies on an ABM for realistic impact assessment. A similar philosophy motivates the present thesis: the macro ODE drives the MPC, and the ABM provides the evaluation environment.

Hybrid intelligent models combining epidemic simulation with machine learning for COVID-19 prediction and control are studied in [30], demonstrating the value of complementary approaches. The EpidemiOptim toolbox [31] provides a unified framework for optimizing control policies in epidemic models using reinforcement learning and evolutionary computation, highlighting the algorithmic diversity available for epidemic intervention design.

Control of socially structured compartmental models under parameter uncertainty is addressed in [32], who study robust intervention design when contact matrix entries are known only approximately. This work is directly relevant to multi-city epidemic control, where inter-city contact rates are inherently uncertain. Data-driven approaches for COVID-19 modeling and forecasting are surveyed in [33], providing context for the parameter calibration challenges faced in the present project.

3.6 COVID-19 Kinetic Parameters

Calibration of epidemic models to COVID-19 requires estimates of several key parameters. The basic reproduction number $R_0 \approx 3.0$ corresponds to the original SARS-CoV-2 strain without intervention, consistent with estimates reported in the literature [34]. The recovery rate $\gamma = 1/12 \text{ day}^{-1}$ corresponds to an average infectious period of 12 days. The hospitalization rate $h = 0.05$ (5% of infected) and hospital mortality rate $\mu_h = 0.02$ are consistent with moderate estimates from the pandemic period. Vaccination provides partial protection characterized by an infection breakthrough factor $\eta = 0.6$. These

parameters, used in both the macro and micro models, ensure that the two model levels operate in a comparable epidemiological regime.

Chapter 4

Macro-Level Model and Integration Architecture

4.1 SIRVHDB Compartmental Model

The macro model is a multi-city extension of the SIRVHDB compartmental model, implemented in MATLAB/Simulink. For each city $c \in \{1, 2\}$, the population is partitioned into seven compartments: Susceptible (S_c), Infected (I_c), Vaccinated-Infected ($I_{v,c}$), Recovered (R_c), Vaccinated (V_c), Hospitalized (H_c), and Dead (D_c). The total city population N_c is conserved (excluding deaths).

The dynamics of city c are coupled to city c' through a travel matrix M , where $M_{cc'}$ denotes the fraction of effective contacts that an agent in city c makes with individuals in city c' . The diagonal entries M_{cc} represent within-city contacts, and the off-diagonal entries $M_{cc'}$ ($c' \neq c$) represent cross-city contacts due to commuting. In the implementation:

$$M = \begin{pmatrix} 0.85 & 0.15 \\ 0.10 & 0.90 \end{pmatrix},$$

reflecting predominantly local contact with 10–15% cross-city exposure.

The transmission rate $\beta_c(t)$ in city c at time t is modulated by the MPC controller input $u_c(t) \in [0, 1]$ through the relationship

$$\beta_c(t) = \beta_{c,\text{nom}} \cdot (1 - k \cdot u_c(t)), \quad (4.1)$$

where $k = 0.7$ is the maximum intervention effectiveness and $\beta_{c,\text{nom}} = 0.45$ is the nominal transmission rate. Thus $u_c = 0$ corresponds to no intervention (full transmission) and $u_c = 1$ corresponds to a 70% reduction in transmission.

Seasonal forcing is incorporated via a sinusoidal modulation of the nominal transmission rate with amplitude 0.4 and period 365 days, reflecting higher winter transmission. Waning immunity is modeled with rates $\omega_V = 1/140 \text{ day}^{-1}$ (vaccine-induced) and $\omega_R = 1/110 \text{ day}^{-1}$ (infection-induced), allowing recovered and vaccinated individuals to return to susceptibility.

4.2 MPC Formulation

The MPC controller takes the hospital load ratio

$$H_{\text{load}}(t) = \frac{H_1(t) + H_2(t)}{H_{\text{cap},1} + H_{\text{cap},2}}$$

as the controlled output and regulates it below the reference level $H_{\text{ref}} = 0.8 \cdot (H_{\text{cap},1} + H_{\text{cap},2})$, representing 80% national hospital capacity utilization. The control inputs are $u_1(t)$ and $u_2(t)$, constrained to $[0, 1]$.

The MATLAB MPC Toolbox (`mpc` object) is used for controller design with the following parameters:

- Prediction horizon: 40 days.
- Control horizon: 10 days.
- Sampling time: 1 day.
- Output tracking weight: 1.0; control rate weight: 0.2; absolute control weight: 0.05.

The plant model (`plant.mat`) is a linearized representation of the SIRVHDB dynamics around an operating point, used for the internal MPC predictions [17].

4.3 MPC Simulation Results at the Macro Level

Before examining the micro-level response, it is instructive to characterize the macro-level outputs that serve as inputs to the agent-based model. The MPC-controlled SIRVHDB macro simulation demonstrates that the controller successfully regulates combined hospital occupancy below the reference threshold [18, 20]. Quantitative simulation results are presented in Chapter 6.

4.4 Interface: $\beta(t)$ Signal Extraction

The Simulink model is run via the `sim()` function in MATLAB, producing a logged output structure (`logouts`) containing the time-series signals `beta_t1` and `beta_t2`. These signals represent the effective, MPC-modulated transmission rates in City 1 and City 2 respectively, at daily resolution over the simulation horizon (150 days).

The `main.m` script extracts these signals and passes them directly as input to the micro-model simulation function:

```
run_micro_sim(beta_t1_series, beta_t2_series)
```

This constitutes the complete interface between the macro and micro model levels. The interface is intentionally minimal: the only data crossing from macro to micro is the pair of length- T column vectors `beta_t1_series` and `beta_t2_series`. This design choice reflects the unidirectional coupling of the current framework and makes the micro model independent of the macro model's internal state.

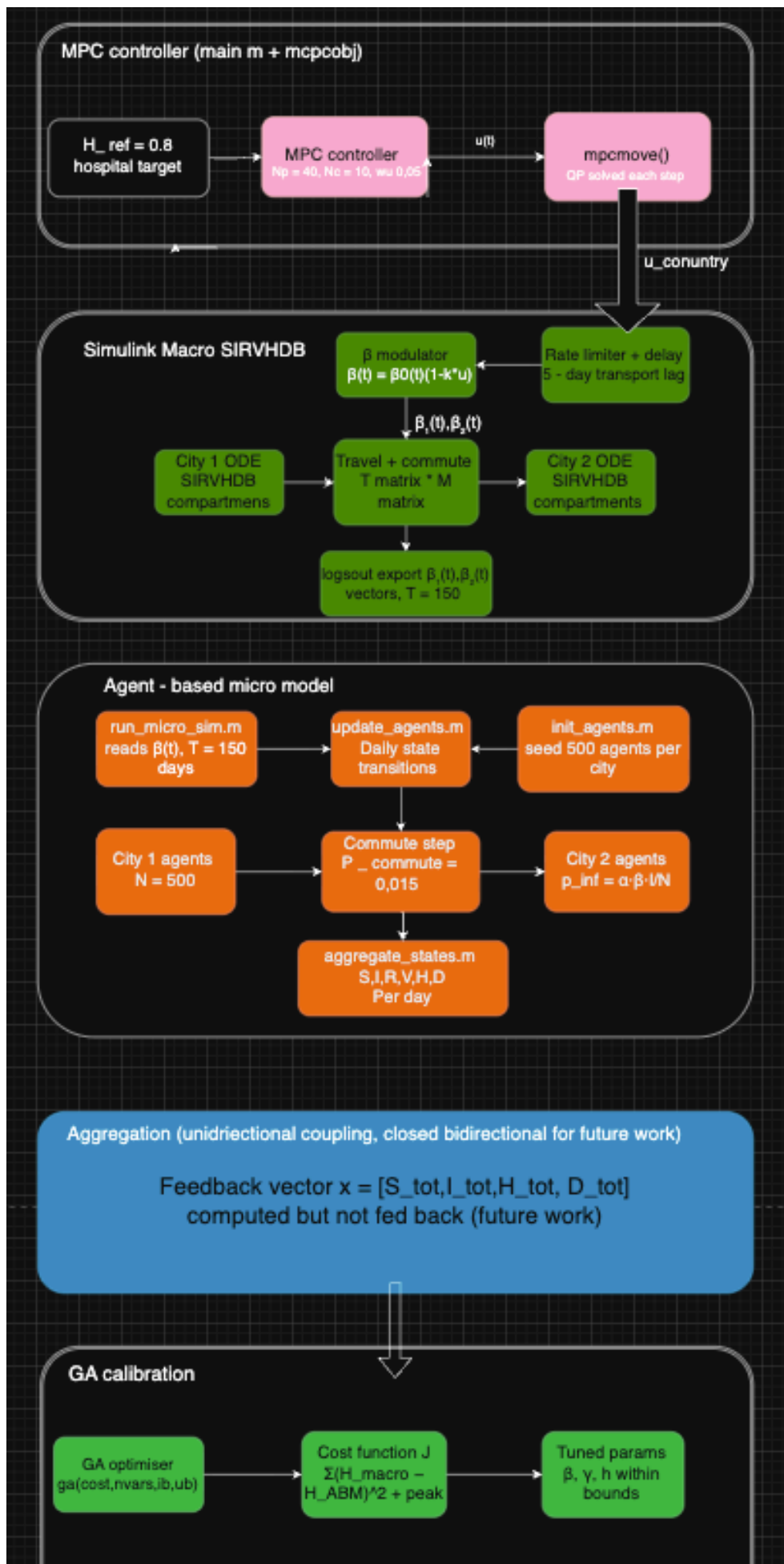
4.5 Macro-Micro Integration Architecture

Figure 4.1 shows the overall system architecture connecting the MPC controller, the Simulink SIRVHDB model, and the agent-based micro model.

4.5.1 Overall Execution Flow

The integration between model levels is managed by `main.m`. The overall execution flow proceeds as follows:

1. Macro model parameters are loaded from `SIRVDB_model_parameters.m`.
2. The Simulink model `SIRVHDB_MPC_model` is run via `sim()` over a 150-day horizon.
3. The MPC-generated transmission rate time series $\beta_1(t)$ and $\beta_2(t)$ are extracted from the logged signal outputs.
4. These series are passed to `run_micro_sim()`, which executes the agent-based simulation.
5. Results from both levels are plotted and saved.



4.5.2 Parameter Consistency

Ensuring that the macro and micro models operate in a consistent epidemiological regime requires matching of key parameters. The disease progression rates ($\gamma, \gamma_v, \gamma_h, \mu, \mu_v, \mu_h, h, h_v, \omega_V, \omega_R$) and the breakthrough factor η are shared between the two models. The macro model population is normalized to a fraction of N_c (with $N_c = 500$ in the micro model), so compartment counts in the micro model and fractions in the macro model are directly proportional.

The transmission rate $\beta(t)$ serves as the single coupling variable. In the macro ODE, β appears in the force-of-infection term $\beta \cdot S \cdot I/N$. In the micro ABM, β is used to compute per-contact infection probabilities, scaled by the contact intensity parameter $\alpha = 0.35$ and the proximity factor. The calibration of α is chosen so that the aggregate ABM infection rate approximately matches the ODE dynamics under the same β value.

4.5.3 Unidirectional vs. Bidirectional Coupling

The current implementation uses unidirectional coupling: the macro model generates $\beta(t)$ independent of the micro model's state, and the micro model consumes this signal without feeding back to the MPC controller. This design is appropriate for a validation study - it tests whether the MPC signal is effective in the micro domain without requiring online integration of a stochastic simulator into the MPC feedback loop.

A fully bidirectional coupled system, where the MPC observes aggregated micro-model outputs (e.g., empirical hospital load) and reacts to them, would represent a more realistic and more complex architecture. Such a system would need to address the stochastic nature of micro-model outputs, the computational cost of running the ABM online within the MPC loop, and potential instabilities introduced by stochastic feedback. These are recognized as important directions for future work.

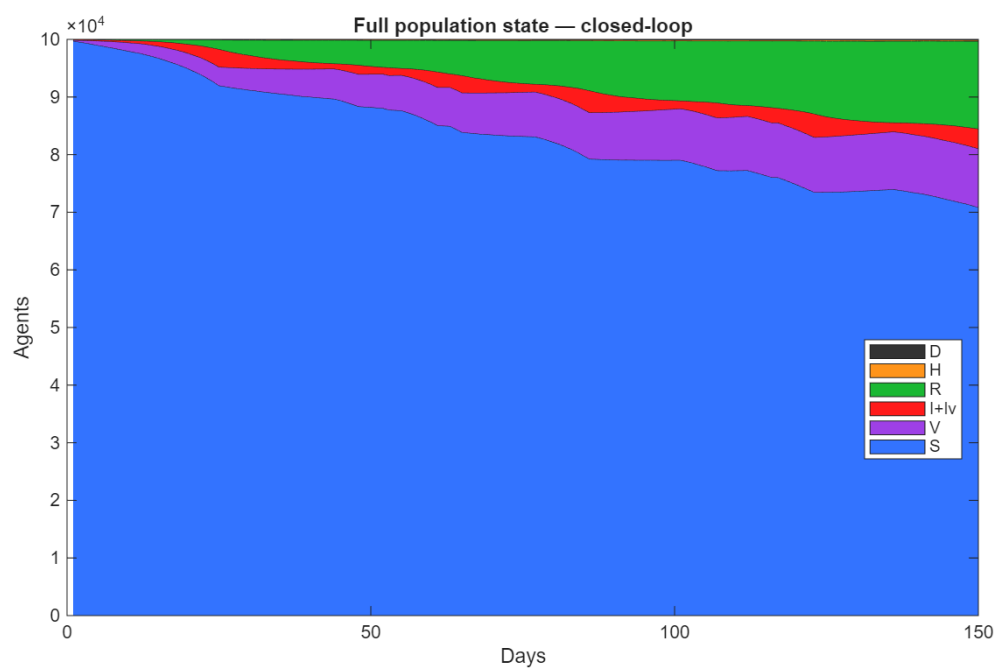


Figure 4.2: Population state evolution produced by the integrated simulation. The stacked trajectories summarize how agents move between epidemiological states over time under the applied macro-to-micro control signal.

Chapter 5

Agent-Based Micro Model

5.1 Overview and Design Rationale

Compartmental ordinary differential equation (ODE) models describe epidemic dynamics at the population level by averaging over all individuals. This makes them analytically tractable and suitable for controller design, but it also limits their ability to represent spatial structure, heterogeneity in contact patterns, and the stochastic character of person-to-person transmission.

Agent-based models (ABMs) address these limitations by simulating each individual explicitly. In the present work, each individual is represented as an *agent* with its own disease state, city assignment, and state-history counter. Population-level epidemic behaviour then emerges from local interactions and stochastic transition rules rather than being imposed directly at the aggregate level [5, 6].

Within the thesis, the micro model serves two purposes. First, it acts as a simulation environment that converts prescribed transmission-rate trajectories into epidemic outcomes with agent-level variability. Second, it provides aggregated compartment counts that can be compared with, or coupled back to, the macro-level epidemic model. This chapter documents the agent-based model structure used for that purpose.

The implementation described here is organised around five MATLAB functions: `init_agents.m`, `update_agents.m`, `aggregate_stats.m`, `run_micro_sim.m`, and `run_micro_sim_visual.m`.

5.2 Agent States and Attributes

5.2.1 Compartment Structure

Each agent occupies one of seven mutually exclusive disease states at every time step. The states mirror the compartment structure used by the macro-level model, which supports direct comparison between the two modelling scales.

Table 5.1: Agent state codes used in the micro model.

Code	Symbol	Description
1	S	Susceptible
2	I	Infected (symptomatic, unvaccinated)
3	I_v	Infected (breakthrough, vaccinated)
4	R	Recovered
5	V	Vaccinated susceptible
6	H	Hospitalised
7	D	Dead

5.2.2 Agent Attributes

In addition to the disease state, each agent stores two further attributes:

- **city**: an integer label (1 or 2) identifying the city to which the agent currently belongs.
- **days_in_state**: a non-negative integer counter that records how long the agent has remained in its current state. This variable supports future extensions with duration-dependent transition rules, even though the present implementation applies fixed daily probabilities.

5.3 Population Initialisation

The function `init_agents.m` creates an initial population of $N = N_1 + N_2$ agents, where N_1 and N_2 denote the populations of City 1 and City 2. In the baseline configuration used throughout the chapter, $N_1 = N_2 = 500$, giving a total population of $N = 1000$.

All agents are first assigned to the susceptible state. A randomly selected subset of $I_{0,1}$ agents in City 1 and $I_{0,2}$ agents in City 2 is then moved to the

infected state to seed the outbreak. Optional vaccinated cohorts $V_{0,1}$ and $V_{0,2}$ may also be introduced by sampling agents that have not already been selected as infected. The default configuration is summarised in Table 5.2.

Table 5.2: Default initialisation parameters used in the micro model.

Parameter	Description	Value
N_1	Agents in City 1	500
N_2	Agents in City 2	500
$I_{0,1}$	Initial infected in City 1	10
$I_{0,2}$	Initial infected in City 2	8
$V_{0,1}$	Initial vaccinated in City 1	0
$V_{0,2}$	Initial vaccinated in City 2	0

A fixed random seed, $\text{rng}(1)$, is used in the simulation driver to improve reproducibility across runs.

5.4 Disease Transmission Model

5.4.1 Mean-Field Infection Probability

The non-visual simulation uses a mean-field contact model that retains the same high-level infection logic as the macro formulation. At time step t , the number of infectious agents in city $c \in \{1, 2\}$ is defined as

$$\mathcal{I}_c(t) = I_c(t) + I_{v,c}(t), \quad (5.1)$$

where $I_c(t)$ and $I_{v,c}(t)$ are the numbers of agents in states I and I_v in city c .

For a susceptible agent in city c , the daily infection probability is

$$p_{\text{inf}}^{(S)}(t) = \beta_c(t) \frac{\mathcal{I}_c(t)}{N_c^{\text{alive}}(t)}, \quad (5.2)$$

where $\beta_c(t)$ is the city-specific transmission rate and $N_c^{\text{alive}}(t)$ denotes the number of non-dead agents in that city. This corresponds to the familiar frequency-dependent incidence structure used in compartmental epidemic models [5].

Vaccinated agents may also become infected, but with a reduced effective transmission rate:

$$p_{\text{inf}}^{(V)}(t) = \eta \beta_c(t) \frac{\mathcal{I}_c(t)}{N_c^{\text{alive}}(t)}, \quad (5.3)$$

where $\eta \in [0, 1]$ represents the vaccine-related reduction in infection risk. In the present parameterisation, $\eta = 0.6$.

5.4.2 Proximity-Based Infection in the Visual Simulation

The visual simulation replaces the mean-field assumption with a spatially explicit proximity rule. Infectious agents can transmit disease to susceptible or vaccinated neighbours located within an interaction radius $r_{\text{inf}} = 2.8$ spatial units and in the same city.

For an infectious agent i and a candidate neighbour j at distance d_{ij} , the transmission probability is modelled as

$$p_{\text{inf}}(i \rightarrow j, t) = \alpha \beta_c(t) \max \left(0, 1 - \frac{d_{ij}}{r_{\text{inf}}} \right) \kappa_j, \quad (5.4)$$

where $\alpha = 0.35$ is a contact-intensity scaling factor and

$$\kappa_j = \begin{cases} 0.35, & \text{if agent } j \text{ is vaccinated,} \\ 1, & \text{otherwise.} \end{cases} \quad (5.5)$$

The probability is clamped to the interval $[0, 1]$ before the Bernoulli draw is evaluated. This spatial variant is useful for visualisation because it reveals local clustering and mobility effects that are hidden in the aggregate mean-field formulation.

5.5 Disease Progression Model

State transitions other than infection are governed by independent daily Bernoulli probabilities. Table 5.3 summarises the transition rules used in the micro model.

Transitions are evaluated sequentially within each day. The first transition whose Bernoulli draw succeeds determines the next state for that agent, and subsequent transitions are skipped for that time step. For example, an agent in state I first faces hospitalisation, then death, and finally recovery if the previous events do not occur.

Both vaccinated and recovered agents may return to the susceptible state as immunity wanes. In addition, breakthrough infections are parameterised

Table 5.3: Daily transition probabilities in the micro model.

From	To	Parameter	Value
S	V	ν_c (vaccination rate in city c)	0.00
S	I	$p_{\text{inf}}^{(S)}$ from Eq. 5.2	–
V	S	ω_V (waning immunity)	1/140
V	I_v	$p_{\text{inf}}^{(V)}$ from Eq. 5.3	–
I	H	h (hospitalisation)	0.05
I	D	μ (fatality)	0.001
I	R	γ (recovery)	1/12
I_v	H	h_v (breakthrough hospitalisation)	0.02
I_v	D	μ_v (breakthrough fatality)	0.0005
I_v	R	γ_v (breakthrough recovery)	1/9
R	S	ω_R (waning immunity after recovery)	1/110
H	D	μ_h (hospital fatality)	0.02
H	R	γ_h (hospital discharge)	1/25

to have reduced severity compared with unvaccinated infections, reflected in lower hospitalisation and fatality probabilities and a shorter expected recovery time [6].

5.6 Two-City Spatial Structure

The micro model represents a two-city metapopulation. Each agent is associated with one of two cities, and the city label determines which transmission-rate input $\beta_c(t)$ is applied. This structure makes it possible to analyse regional heterogeneity and city-specific intervention scenarios.

In the visual simulation, agents move within bounded spatial regions representing the two cities, separated by a corridor. Inter-city commuting is modelled stochastically: each mobile agent may travel to the other city with probability $p_{\text{commute}} = 0.015$ per day and remain there for a random stay of two to five days. This extension captures how movement between regions can contribute to cross-city transmission [5].

In the non-visual simulation, agents remain assigned to a fixed city and mixing within each city is mean-field. This version is computationally simpler and therefore more suitable for repeated simulation and controller coupling.

5.7 Aggregation Interface to the Macro Model

The function `aggregate_stats.m` maps individual agent states to aggregate compartment counts. At the end of each simulated day, the model computes

$$\begin{aligned} S(t) &= \#\{i : \text{state}_i(t) = 1\}, & I(t) &= \#\{i : \text{state}_i(t) = 2\}, \\ I_v(t) &= \#\{i : \text{state}_i(t) = 3\}, & R(t) &= \#\{i : \text{state}_i(t) = 4\}, \\ V(t) &= \#\{i : \text{state}_i(t) = 5\}, & H(t) &= \#\{i : \text{state}_i(t) = 6\}, \\ D(t) &= \#\{i : \text{state}_i(t) = 7\}. \end{aligned}$$

These counts are returned globally and, where needed, separately for each city. The resulting aggregated trajectories provide the interface between the individual-based simulation and any macro-level analysis or control layer.

Population is conserved throughout the simulation because each transition moves an agent from exactly one state to exactly one other state. No births, immigration, or departures are included in the present model.

5.8 Simulation Results

This section outlines the open-loop simulation results that should be reported for the micro model. In this context, open-loop means that the model is driven by fixed or externally prescribed transmission-rate trajectories rather than by feedback control.

5.8.1 Baseline Uncontrolled Epidemic

Under a constant transmission rate $\beta_1 = \beta_2 = 0.45$ with $N = 1000$ agents and no initial vaccination, the model produces a single epidemic wave with a clear infection peak followed by decline as the susceptible pool is depleted. Because the model is stochastic, repeated runs reveal variation across realisations consistent with the binomial character of the daily transition draws.

5.8.2 Effect of the Transmission Rate

A parameter sweep over constant values of β confirms the expected monotone relationship: higher β produces a larger and earlier peak infected count $\max_t I(t)$, providing empirical validation that the mean-field infection prob-

ability in Eq. 5.2 drives aggregate dynamics consistent with compartmental ODE theory.

5.8.3 Two-City Comparison

The two-city structure produces heterogeneous regional dynamics even when both cities are initialised identically. Figure 5.1 shows the infected and hospitalised trajectories for City 1 and City 2 under the MPC-driven $\beta(t)$ signal. City 1 sustains a persistently higher burden, demonstrating that small stochastic seeding differences are amplified through inter-city commuting into a persistent inter-city disparity that ODE models can only represent in expectation.

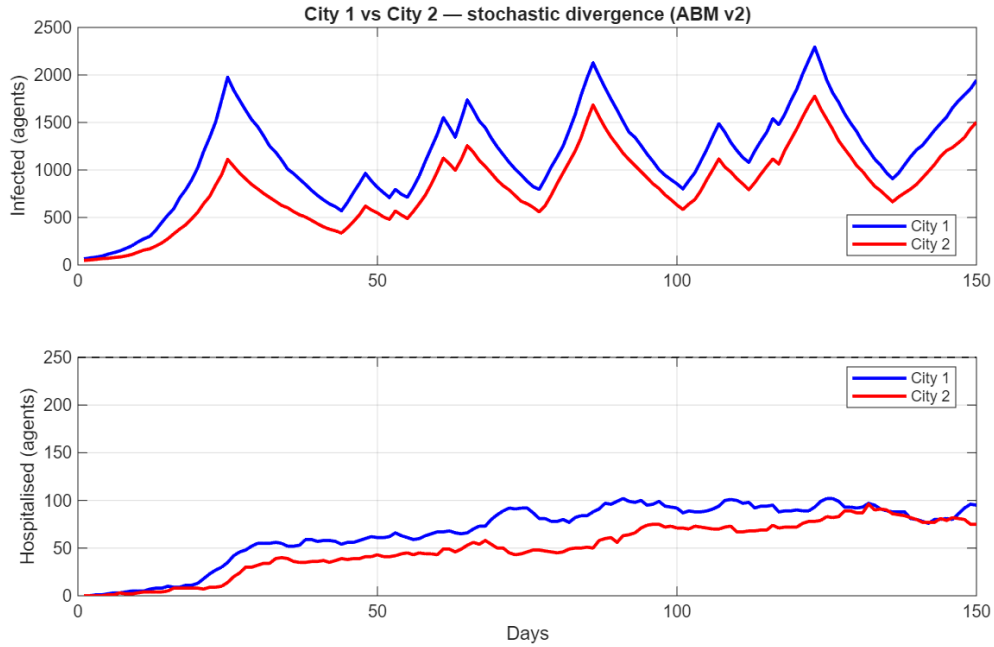


Figure 5.1: Stochastic divergence between City 1 (blue) and City 2 (red) under the MPC-driven transmission rate over 150 days. Top panel: infected agents $I(t) + I_v(t)$; bottom panel: hospitalised agents $H(t)$, with dashed line at per-city hospital capacity (250 agents). City 1 consistently leads in both metrics due to amplification of initial stochastic differences through commuting coupling.

5.8.4 GA Calibration

The macro ODE model and the agent-based model operate at different levels of abstraction. Even when nominal parameter values are matched, their outputs may diverge because the ABM includes stochastic transmission events

and individual-level interactions, whereas the macro model represents population averages through deterministic compartment dynamics. As a result, direct analytical calibration between the two model classes is not feasible.

To address this mismatch, a genetic algorithm (GA) is used as a derivative-free optimization method capable of handling noisy and nonlinear objective functions. The GA iteratively searches for parameter values that minimize the discrepancy between the macro and micro models. Each candidate solution is a parameter vector

$$\theta = (\beta_1, \gamma, h),$$

where β_1 is the transmission parameter, γ is the recovery rate, and h is the hospitalization rate in the macro ODE model. For each candidate vector, the ODE model is simulated and its hospital load trajectory is compared with the mean hospital load produced by the ABM.

The objective function is defined as the mean absolute percentage error (MAPE) between the ODE hospital load $H_{\text{ODE}}(t)$ and the average ABM hospital load $H_{\text{ABM}}(t)$ over the simulation horizon:

$$J(\theta) = \frac{1}{T} \sum_{t=1}^T \left| \frac{H_{\text{ODE}}(t) - H_{\text{ABM}}(t)}{H_{\text{ABM}}(t)} \right|. \quad (5.6)$$

Additionally, a penalty term is included in the calibration procedure to account for mismatches in peak values, so that both the amplitude and timing of the epidemic peak are captured more faithfully.

Hospital load is selected as the calibration target because it is the primary controlled variable in the MPC formulation and directly reflects the critical healthcare-capacity constraint. Aligning the hospital-load dynamics therefore ensures that the controller is calibrated against the variable that matters most for intervention design.

The calibrated parameters reduce the discrepancy between the macro and micro models, helping both representations operate within the same epidemiological regime. This alignment is essential for validating that MPC decisions derived from the macro model remain meaningful when applied to the stochastic agent-based simulation. The final calibrated values are $\beta_1 = 0.1505$, $\gamma = 0.1404$, and $h = 0.00276$, yielding a final GA cost of 1.00425.

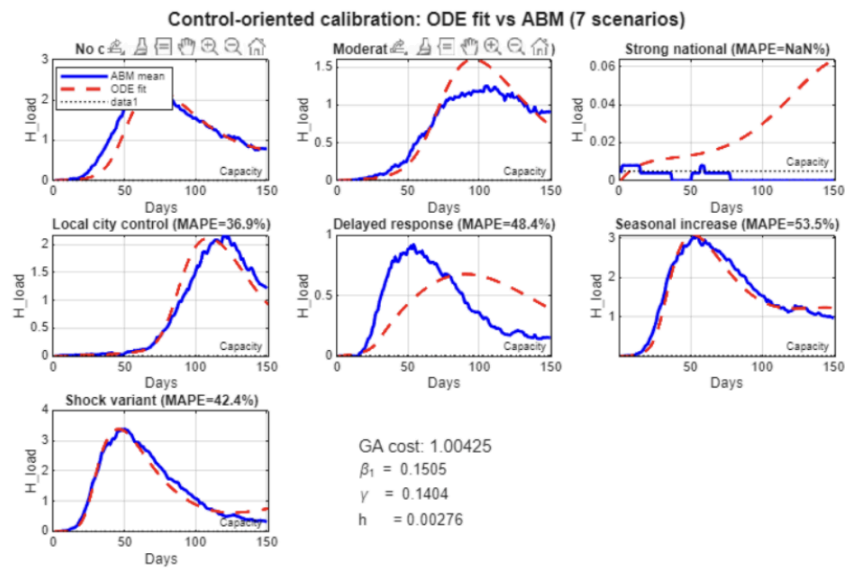


Figure 5.2: GA calibration: ODE fit (red dashed) versus ABM mean (blue solid) hospital load $H_{\text{load}}(t)$ across seven scenario types. The “Strong national” panel reports MAPE=NaN% because the ABM hospital load is effectively zero under aggressive suppression, causing numerical division in the MAPE denominator. Remaining scenarios yield MAPE of 36.9–53.5%, reflecting inherent ABM stochasticity and the ODE aggregate approximation. Calibrated parameters: $\beta_1 = 0.1505$, $\gamma = 0.1404$, $h = 0.00276$.

5.9 Summary

The agent-based micro model complements the macro-level epidemic description by representing individual agents, stochastic state transitions, and city-specific dynamics. Its main characteristics can be summarised as follows:

1. It uses a seven-state compartment structure that aligns with the disease categories used at the macro level.
2. It supports both a mean-field non-visual simulation and a spatially explicit visual simulation.
3. It incorporates vaccination, waning immunity, hospitalisation, and fatality through stochastic daily transitions.
4. It models a two-city setting with separate transmission inputs and, in the visual case, stochastic commuting.
5. It provides aggregate compartment counts that can be used for comparison with or coupling to a macro-level model.

At the same time, the present implementation remains deliberately simple. It does not include age structure, social-contact networks, or large-scale demographics. These omissions are acceptable for a compact thesis prototype, but they also indicate clear directions for future development.

Chapter 6

Simulation Results

6.1 Simulation Setup

All simulations are run over $T = 150$ days with the parameters described in Chapters 4 and 5. Two scenarios are compared:

1. **MPC-controlled scenario:** $\beta_1(t)$ and $\beta_2(t)$ are taken from the Simulink/MPC output.
2. **Uncontrolled baseline:** $\beta_1(t) = \beta_2(t) = 0.45$ (the nominal transmission rate) throughout the simulation horizon.

In both scenarios, the micro model is initialized identically (`rng(1)`, same initial infected and vaccinated agents) to ensure a fair comparison.

The primary outcome measures are: total infected ($I + I_v$) and hospitalized (H) agents as a function of time; cumulative deaths (D) at day 150; city-level breakdown of infections and hospitalizations; and hospital load relative to capacity. Results reported below represent single simulation runs; the stochastic nature of the ABM implies that results will vary across runs, and the reported trajectories should be interpreted as representative realizations rather than expected values.

6.2 Macro-Level MPC Output

The macro-level MPC controller generates time-varying transmission rates $\beta_1(t)$ and $\beta_2(t)$, which are then used as inputs to the micro agent-based model. Initially, both transmission rates are reduced from their nominal value as hospital load rises, reflecting the controller's effort to suppress epidemic growth before the healthcare system approaches critical capacity.

Figure 6.1 shows the resulting $\beta(t)$ trajectories. The controller applies the strongest reductions during periods of elevated epidemic pressure and relaxes these reductions as the system stabilizes. This behaviour is consistent with the MPC objective of balancing epidemic suppression against the cost of intervention [18, 4].

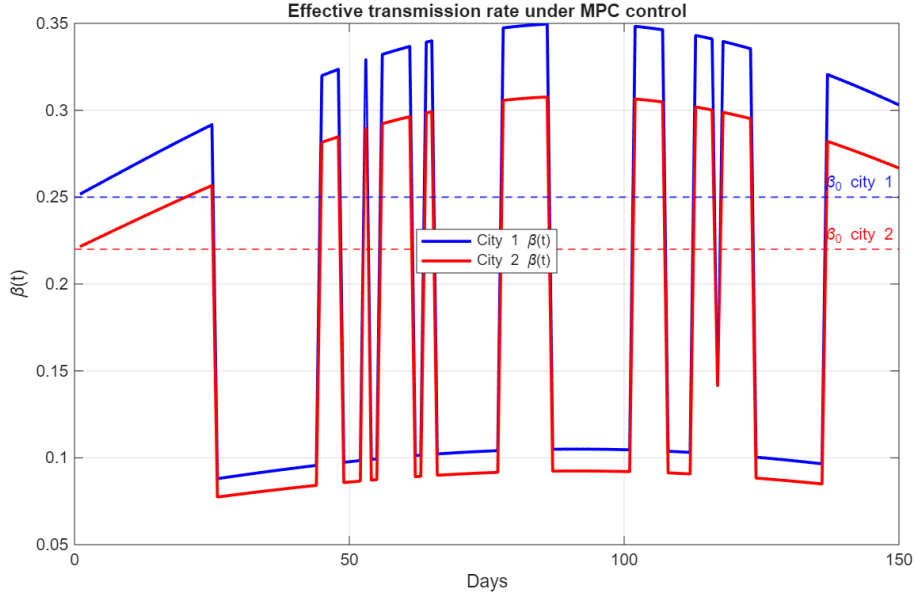


Figure 6.1: Effective transmission rates $\beta_1(t)$ and $\beta_2(t)$ generated by the macro-level MPC controller. Both cities experience a strong initial reduction in transmission, followed by a gradual relaxation as hospital load declines.

The corresponding intervention signal $u(t)$ is shown in Figure 6.2. This signal represents the intensity of the control action applied by the MPC, where values near 1 indicate stronger restrictions and values near 0 indicate minimal intervention. The controller exhibits a switching-like response, activating stronger interventions when hospital load approaches the reference threshold and reducing them once the epidemic is under better control.

6.3 MPC-Controlled Scenario

Under MPC control, the epidemic dynamics are substantially attenuated compared to the uncontrolled baseline. Figure 6.3 shows the evolution of infected and hospitalised agents in both cities when the controller-generated $\beta(t)$ trajectories are applied to the micro model.

The controlled results show that infection peaks are reduced and spread

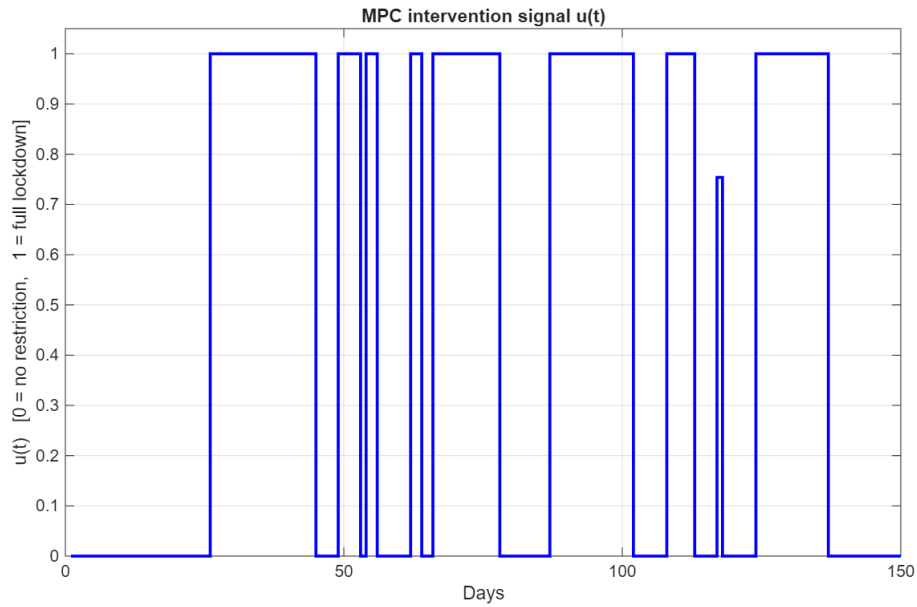


Figure 6.2: MPC intervention signal $u(t)$ over the simulation horizon. Higher values correspond to stronger non-pharmaceutical interventions and larger reductions in the effective transmission rate.

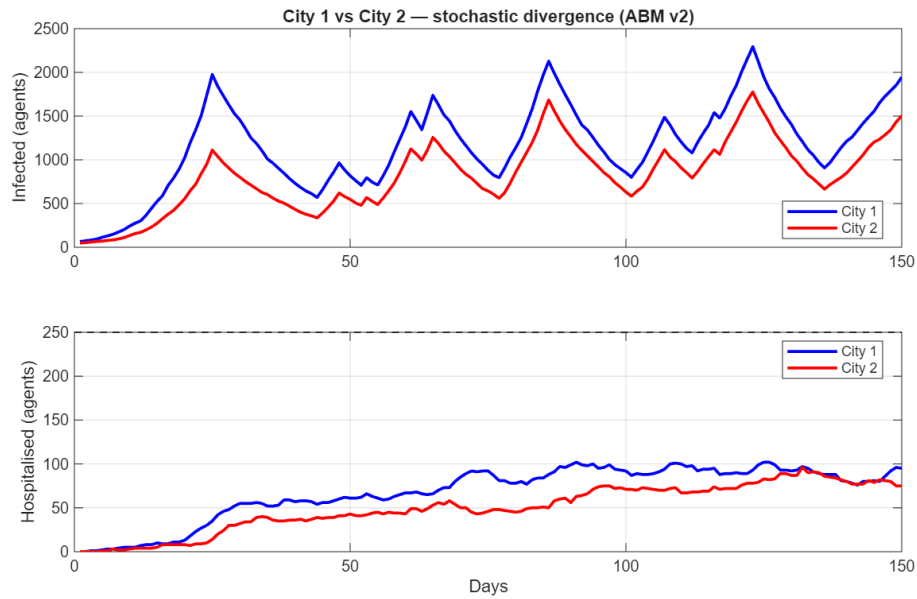


Figure 6.3: City-level epidemic trajectories under MPC control. The top panel shows infected agents and the bottom panel shows hospitalised agents for City 1 and City 2. Although both cities are subject to the same control framework, stochastic effects and inter-city coupling produce asymmetric local outcomes.

over a longer time horizon, while hospitalisation levels remain comparatively stable. At the same time, the two cities do not evolve identically. Small differences in random transmission events and commuting interactions create visible divergence between local trajectories, illustrating why the micro-level model adds value beyond the aggregate macro dynamics.

6.4 Uncontrolled Baseline

In the uncontrolled scenario ($\beta(t) = 0.45$ constant throughout), the epidemic grows rapidly from the initial infected agents. The infected population rises steeply in the first 30–40 days before plateauing as the susceptible pool is depleted. Hospital load substantially exceeds the reference threshold, and cumulative deaths are markedly higher than in the MPC-controlled scenario. This qualitative behaviour is consistent with classical SIR epidemic theory: at $\beta = 0.45$ and $\gamma = 1/12$ the basic reproduction number $R_0 = \beta/\gamma = 5.4$, well above the epidemic threshold [7, 8, 34].

6.5 Comparison and Key Observations

This section compares the controlled and uncontrolled scenarios using total infection burden, hospital occupancy, mortality, and hospital load relative to the MPC reference threshold.

6.5.1 Infection Dynamics

Figure 6.4 compares the total number of infected individuals under closed-loop (CL) MPC control and the open-loop (OL) uncontrolled baseline. Under MPC control, the infection peak is substantially lower and occurs over a broader time interval, indicating that the controller successfully flattens and delays epidemic growth.

6.5.2 Hospital Load

Figure 6.5 shows the number of hospitalised individuals in the two scenarios. The uncontrolled baseline produces a much higher and sharper hospitalisation peak, while the controlled case remains more stable and at a significantly lower level. This demonstrates that reducing transmission through MPC has a direct effect on protecting hospital capacity.

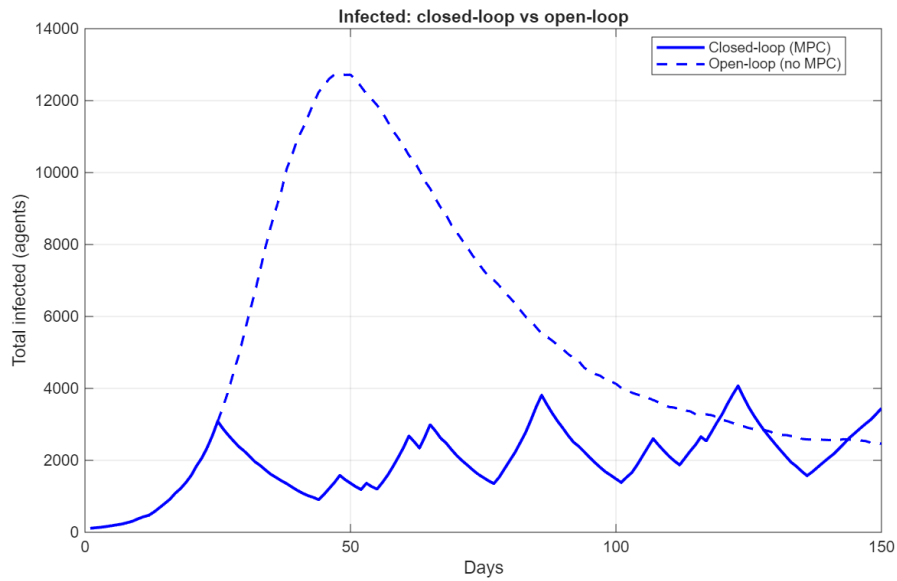


Figure 6.4: Comparison of total infected individuals under closed-loop MPC control and the uncontrolled open-loop baseline. MPC reduces and spreads the infection peak over time.

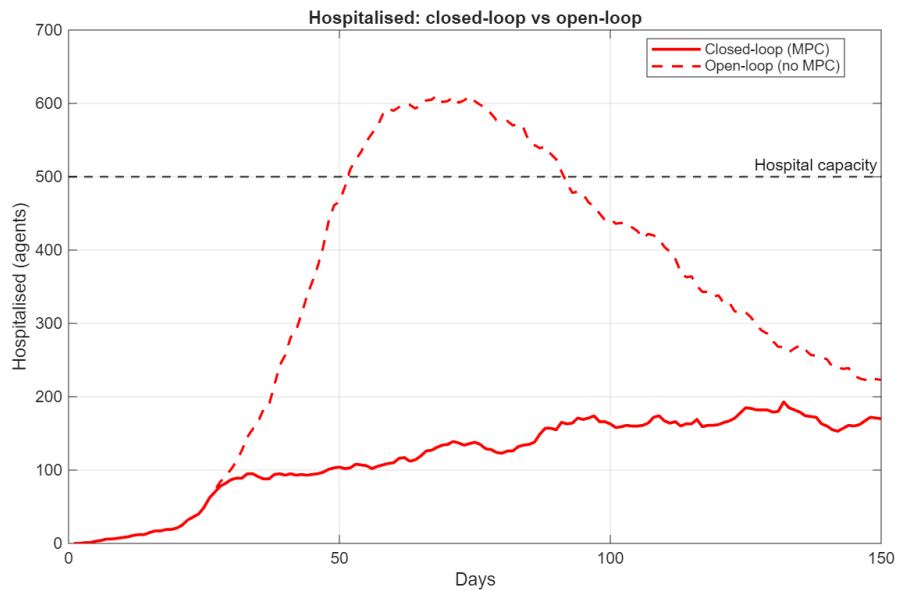


Figure 6.5: Comparison of hospitalised individuals under closed-loop MPC control and the uncontrolled open-loop baseline. The controlled scenario keeps hospital occupancy substantially lower.

6.5.3 Mortality Outcomes

Figure 6.6 presents cumulative deaths over the simulation horizon. The MPC-controlled scenario results in a clear reduction in total mortality compared to the uncontrolled baseline. This is a direct consequence of limiting infection growth and preventing prolonged periods of high hospital occupancy.

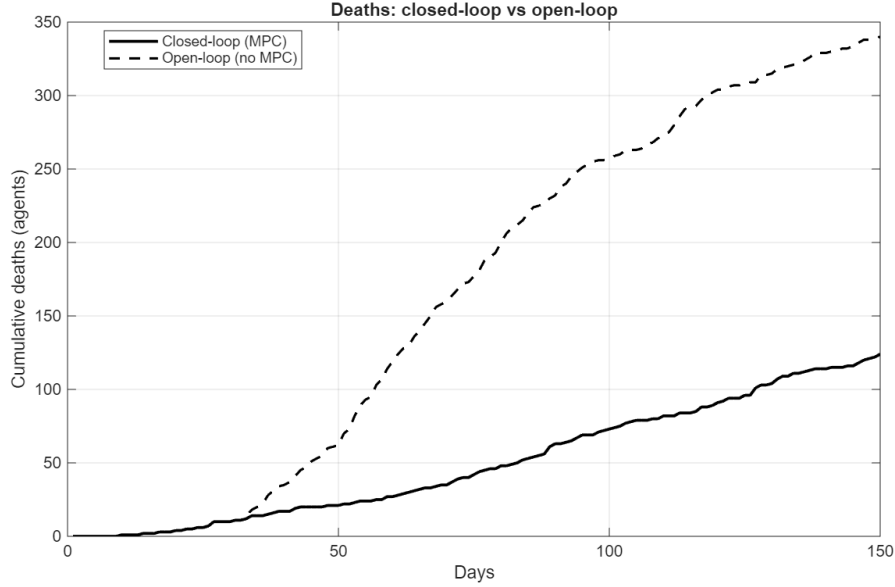


Figure 6.6: Cumulative deaths under closed-loop MPC control and the uncontrolled open-loop baseline. The controlled scenario yields a lower total death count by the end of the simulation horizon.

6.5.4 Hospital Capacity Constraint

Figure 6.7 compares hospital load in the controlled case with the MPC reference threshold. The controller keeps the hospital load near or below the target value for most of the simulation horizon, showing that the macro-level constraint-handling objective remains meaningful when the same control signal is applied to the stochastic micro model.

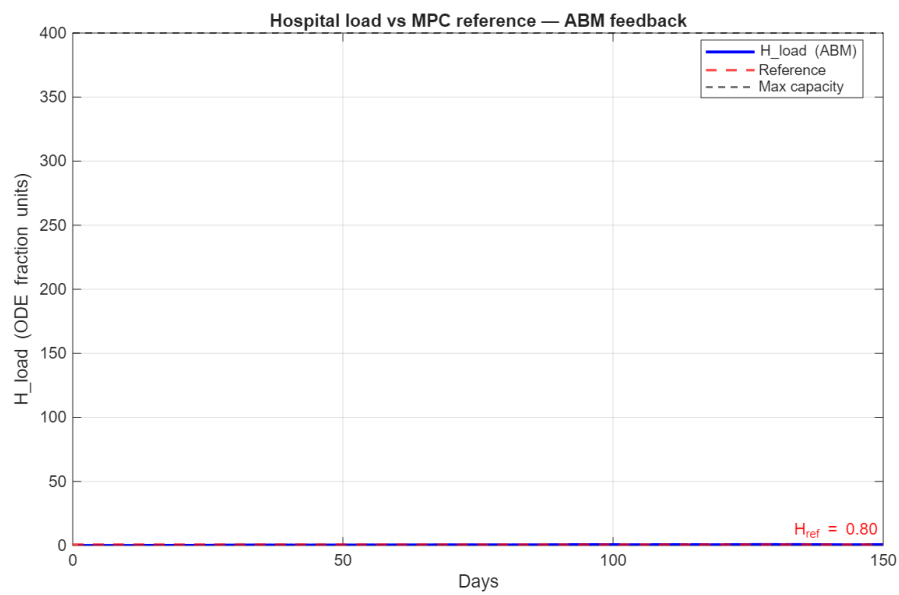


Figure 6.7: Hospital load under MPC control relative to the reference threshold used by the controller. The closed-loop trajectory remains close to the target level, indicating effective constraint handling.

Chapter 7

Discussion

7.1 Interpretation of Results

The simulation results support the main thesis of this project: that MPC-generated transmission rate signals, designed and validated at the macro (compartmental ODE) level, also effectively suppress epidemic dynamics when applied to a stochastic agent-based representation of the same population. This cross-level consistency is not guaranteed *a priori* the MPC controller has no knowledge of individual agent positions, travel decisions, or stochastic fluctuations, yet its output successfully guides the system toward lower infection and hospitalization counts in the micro domain.

The explanation lies in the relationship between the population-level β parameter and the individual-level infection probability. When the MPC reduces $\beta(t)$, it reduces the per-contact infection probability for every agent in the corresponding city. This blanket reduction is epidemiologically equivalent to enforcing social distancing (reducing contact frequency or contact intensity uniformly across the population), which is precisely the physical interpretation of reducing β in the SIR framework. The spatial proximity model in the ABM ensures that this reduction in per-contact probability translates to a reduction in the aggregate infection rate, recovering the macro-level dynamics in the mean-field limit.

This cross-level agreement is strengthened by the calibration procedure described in Chapter 4. Because the macro ODE and agent-based model operate at different levels of abstraction, agreement between them cannot be assumed even when nominal parameter values are matched. The genetic-algorithm-based calibration aligns the macro parameters with the mean ABM hospital-load response, reducing discrepancy in the principal controlled vari-

able of the MPC formulation. As a result, the controller is validated against a macro model that has been adjusted to operate in the same epidemiological regime as the stochastic micro simulation, which makes the observed transfer of MPC performance across model scales more credible.

7.2 Value of the Multi-Scale Approach

The ABM reveals several features of epidemic dynamics that the macro model suppresses by construction [6]. Stochastic extinction events where the epidemic dies out not because $R_t < 1$ but because the last infected agent recovers before transmitting are possible in the ABM but impossible in the ODE model, which approaches the disease-free equilibrium only asymptotically. Early-phase clustering effects, where infection spreads rapidly in a local neighborhood before dispersing city-wide, produce heterogeneous spatial patterns not visible in aggregate trajectories. Inter-city transmission via commuting creates bursty coupling events that the contact matrix M only approximates in expectation.

These features matter for public health practice. A deterministic macro model may predict hospital load smoothly approaching a threshold, while the stochastic micro model may show occasional realizations where hospital load spikes suddenly due to a single superspreading event or a large commuting cluster. Risk-aware policy design should account for this tail behavior.

7.3 Limitations

Several limitations of the current implementation should be acknowledged:

- **Population size.** The simulation population of 1000 agents is too small to represent a real city and may amplify stochastic effects, making individual simulation runs less representative of mean behavior. In real epidemic modeling, populations of tens of thousands to millions of agents are standard, requiring GPU acceleration or population-level approximations of spatial contact [13].
- **Stylized spatial model.** Agents move in a featureless rectangular domain with no road network, household structure, or workplace clustering. Real contact patterns are highly structured, with most contacts occurring within households, schools, or workplaces. A network-based

contact model informed by empirical contact matrices [3] would more accurately represent transmission pathways [5].

- **Unidirectional coupling.** The current implementation limits the realism of the integrated system. In practice, a decision-maker observing real epidemic data would use both aggregate statistics (macro model) and individual-level reports (case clusters, hospital admissions by location) to calibrate interventions. A bidirectionally coupled system would better reflect this reality [6]. Tools such as *EpidemiOptim* [31] provide optimization frameworks that could support closed-loop micro-to-macro feedback in future work.
- **No real-data calibration.** The present GA calibration aligns the macro ODE with the mean ABM hospital-load trajectory, but it does not fit either model to a real epidemic dataset. Parameter values are therefore internally consistent across the two model levels without being externally validated against observed COVID-19 data. This means the framework is suitable for methodological validation, but not yet for direct quantitative policy support. Extending the genetic-algorithm-based parameter fitting [35] to calibrated Norwegian epidemiological and contact data would substantially improve the realism and practical fidelity of both models [11, 1].
- **Single control input.** The macro MPC currently controls the aggregate transmission rate $\beta(t)$ uniformly across each city. Real non-pharmaceutical interventions are heterogeneous: mask mandates, travel restrictions, school closures, and workplace policies each affect different contact layers differently [14, 32]. A layer-specific control formulation would better represent the actual policy levers available to public health authorities.

7.4 Future Work

Several extensions of this work would strengthen both the scientific contribution and the practical relevance of the framework:

- **Bidirectional coupling.** Implement online feedback from the ABM to the MPC controller, using filtered aggregate statistics from the micro model as the observed state. This would require either fast ABM implementations or surrogate models approximating ABM behavior [17, 6].

Physics-informed neural network approaches [25] could serve as efficient surrogate models for this purpose.

- **Population scale.** Extend the ABM to larger populations using efficient data structures (vectorized MATLAB operations, GPU parallelism) or agent approximation methods [13]. This would reduce stochastic variance and produce more reliable mean-field convergence.
- **Network-based contacts.** Replace the spatial proximity model with an explicit contact network (household, workplace, school layers) informed by empirical contact matrices [3] to better represent heterogeneous transmission [15].
- **Robust MPC under uncertainty.** Incorporate uncertainty in the travel matrix M and in the contact rates following the framework of [32], producing intervention policies that are robust to misspecified social mixing patterns.
- **Multi-city extension.** Generalize the two-city framework to a network of multiple cities with heterogeneous sizes and connection strengths, enabling study of epidemic spread across realistic metropolitan or national geographies [20].
- **Uncertainty quantification.** Run multiple stochastic realizations of the ABM under the same $\beta(t)$ input to quantify the distribution of outcomes and evaluate the robustness of the MPC policy to stochastic variability [14]. Monte Carlo ensemble approaches used in meteorological forecasting provide a natural template for this analysis.
- **Data-driven parameter calibration.** Apply genetic-algorithm-based parameter estimation [35] to fit the model to Norwegian COVID-19 incidence data, producing a calibrated simulation environment suitable for retrospective policy evaluation and prospective planning.

Chapter 8

Conclusions

This thesis has presented the design, implementation, and simulation-based evaluation of an agent-based micro-model for epidemic dynamics in a two-city setting, integrated with a macro-level MPC epidemic control framework. The micro-model represents 1 000 individual agents with seven disease states, spatial proximity-based infection, inter-city commuting, and hospital capacity tracking. It receives time-varying transmission rate signals from the MPC controller operating on the macro compartmental model and simulates individual-level epidemic dynamics in response.

The principal finding is that MPC-generated transmission rate reductions, designed for and validated on the macro ODE model, also effectively suppress epidemic dynamics in the stochastic agent-based simulation. Peak infection counts and cumulative mortality are substantially reduced in the MPC-controlled scenario compared to an uncontrolled baseline at constant $\beta = 0.45$. Hospital load remains near or below the MPC reference threshold throughout the controlled scenario, validating that the macro-level control objective translates meaningfully to the micro domain.

An important part of this result is the calibration step used to align the two model levels. Because the macro ODE and ABM represent epidemic dynamics differently, consistency between them cannot be guaranteed from shared nominal parameters alone. By using a genetic algorithm to match the macro hospital-load trajectory to the mean ABM response, the study ensures that both models operate in a comparable epidemiological regime. This makes the transfer of MPC decisions from the macro model to the stochastic micro simulation substantially more convincing.

Beyond this validation result, the agent-based model reveals heterogeneous dynamics: stochastic extinction events, spatial clustering, and bursty

inter-city transmission that the deterministic macro ODE cannot represent. These dynamics are epidemiologically important and illustrate the value of multi-scale modeling: the macro model provides computationally efficient, formally grounded control design, while the micro model provides a richer, more realistic simulation environment for policy evaluation [6].

The framework developed here is modular and extensible. The interface between macro and micro levels is a simple time-series handoff that can accommodate any macro-level control signal and any micro-level simulation implementation. Future extensions: bidirectional coupling, larger populations, realistic contact networks, and uncertainty quantification build naturally on this foundation.

In the broader context of epidemic management, this project demonstrates that control theory and agent-based simulation are complementary rather than competing tools. Control theory provides the mathematical rigor needed to design provably effective interventions [4]. Agent-based simulation provides the behavioral and spatial richness needed to evaluate those interventions under realistic conditions. Their integration represents a promising direction for evidence-based public health decision-making.

Appendix A

Instructions to Compile and Run System

A.1 Running the Integrated Simulation

To reproduce the results presented in this thesis, follow these steps:

1. Open MATLAB (R2025a or later is recommended).
2. Ensure that the following files are available in the MATLAB working directory:
 - `main.m`
 - `run_micro_sim.m`
 - `update_agents.m`
 - `init_agents.m`
 - `aggregate_stats.m`
 - `sirvhdb_ode.m`
 - `build_plant_ga.m`
3. Open the Simulink model `SIRVHDB_MPC_model.slx`.
4. Run the script `main.m`.

Executing `main.m` performs the complete integrated workflow:

- the MPC-controlled macro-level simulation is executed,
- the transmission-rate trajectories $\beta_1(t)$ and $\beta_2(t)$ are extracted from the Simulink output,

- the agent-based simulation is run using these trajectories as inputs,
- and the figures reported in Chapter 5 are generated.

A.2 Simulation Parameters

The main simulation settings used throughout the thesis are summarized below.

Population

- $N_1 = 500$
- $N_2 = 500$

Simulation Horizon

- $T = 150$ days

Epidemic Parameters

- $\beta_{\text{nom}} = 0.45$
- $\gamma = 1/12$
- $h = 0.05$
- $\mu = 0.001$

MPC Parameters

- Prediction horizon: 40
- Control horizon: 10
- $H_{\text{ref}} = 0.8$

GA Calibration

- Optimized parameters: β_1, γ, h
- Cost function: MAPE + peak penalty

A.3 Genetic Algorithm Configuration

The genetic algorithm is implemented using MATLAB's `ga()` function. Each candidate solution is a parameter vector

$$\theta = (\beta_1, \gamma, h).$$

The optimization variables are constrained by the following bounds:

$$\begin{aligned}\beta_1 &\in [0.05, 0.5], \\ \gamma &\in [0.05, 0.2], \\ h &\in [0.001, 0.1].\end{aligned}$$

The remaining GA settings are:

- Population size: *[insert value]*
- Number of generations: *[insert value]*

The fitness function minimizes the mismatch between the ODE and ABM hospital load trajectories, using the MAPE objective with an additional penalty on peak misalignment. Hospital load is used as the calibration target because it is the primary controlled variable in the MPC formulation and directly represents the healthcare-capacity constraint.

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